

AKSHAYAA D

AI Engineer · Production RAG, Agentic AI & LLM Evaluation

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PROFESSIONAL SUMMARY

3+ years of engineering experience, currently an AI Engineer at FourKites, owning production RAG, LLM, and Agentic AI systems for enterprise supply-chain workflows end-to-end – from early architecture through production reliability – turning ambiguous, high-stakes production problems into measurable scale, reliability, and business-value gains.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Data Structures & Algorithms, Object-Oriented Programming, Low-Level Design (LLD)

LLM & GenAI: Large Language Models (LLM), Transformers, Tokenization, Prompt Engineering, Tool/Function Calling, OpenAI API, Azure OpenAI (GPT-4o), GPT-4 Vision, Fine-Tuning Concepts, Quantization Concepts, KV Cache

RAG & Vector Search: Document Ingestion, Chunking, Hybrid Search (BM25 + Semantic + Reciprocal Rank Fusion), Reranking, Embeddings, Self-Healing RAG, Qdrant, CLIP (Multimodal Embeddings)

Agentic Systems: LangChain, LangGraph, Multi-Agent Systems, Agentic Workflows, MCP (Model Context Protocol)

Automation & Tooling: Browser/Web Automation (Playwright), Pipedream, Claude Code

Backend & Cloud: FastAPI, REST APIs, Async Processing, Distributed Systems, Kafka, Redis, PostgreSQL, ClickHouse, Docker, Kubernetes, AWS/Azure, Azure Blob Storage, Git

Evaluation & Observability: Hybrid Retrieval + LLM-as-Judge Scoring, Agreement-Rate Benchmarking, Observability, MLOps, SigNoz, Chronosphere, Langfuse

PROFESSIONAL EXPERIENCE

Software Engineer II – FourKites, Chennai

June 2023 – Present

Internal transition to AI team, mid-2025

- Led end-to-end ownership of production agentic AI systems—SAM (multimodal RAG on LangChain/FastAPI/Kubernetes, CLIP + Qdrant retrieval, **1.39M+ executions**) and ALAN (AI booking agent on Pipedream/Playwright/Azure OpenAI vision)—demoing directly to the CEO.
- Replaced SAM's hardcoded pipeline with a format-agnostic system (21+ templates, two-stage Azure OpenAI pipeline), fixing a multi-sheet bug that dropped 90%+ of file data and 28,693 garbage records while scaling **2.3x** at 0.1–0.3% failure.
- Root-caused a **3.8x** excess in duplicate shipment creation to a write-before-read idempotency race across Kafka redeliveries and shipped a targeted fix, achieving a **99.6% reduction**—classified as P1 Avoided.
- Owned ALAN's reliability across retailer and 3PL scheduling portal retiring a 428-entry hardcoded site registry in favor of recorded templates served at runtime, turning portal onboarding from an engineering task into a recording session—and cutting the booking failure rate from **60.7% to 8.9%** against a validated **~\$400K annual value**.
- Processed ambiguous, multilingual requirements—including Spanish-language email intake across 10,000+ messages—to onboard a new enterprise customer, delivering Phase 2 in 2 months; recognized by CTOs on both sides.
- Designed ALAN's failure-recovery layer: two-level retry (in-process exponential backoff with jitter and a non-retryable error class, plus a Redis-backed workflow queue), Redis-based session reuse of authenticated browser state with dual idle/lifetime expiry, and a 3-hourly validation cron reconciling appointment state against the portal—catching bookings that reported success but were never actually confirmed.

PERSONAL PROJECTS

Resume Reality Check – Resume/JD Evidence Verifier

FastAPI, RAG, RRF, Evaluation, Gemini, ONNX, BM25, Docker

- Built a hybrid-retrieval (BM25 + semantic + Reciprocal Rank Fusion) system with an LLM-as-judge scored against a fixed rubric, separating what a resume claims from what it actually proves with cited evidence.
- Found and fixed **13 real production issues** – retrieval corpus pollution, non-determinism from LLM serving-layer routing, memory-constrained deployment failure – validated against a 17-pair hand-labeled eval set across 3 different resumes.

Swarm Onboard – Multi-Agent Repo Onboarding Assistant

Python, FastAPI, LangGraph, MCP, Redis, Pydantic, LangSmith, AST Parsing, GitHub REST API

- Designed a multi-agent system that analyzes unfamiliar GitHub repositories and generates onboarding documentation, using a deterministic graph-building layer plus an LLM-based reasoning layer with cross-agent validation.
- Verified generated documentation against **4 different GitHub repositories** varying in size and structure to confirm output quality and consistency.

EDUCATION

Sri Krishna College of Engineering and Technology

Bachelor of Technology (B.Tech), Information Technology

Jul 2019 – May 2023

CGPA: 9.34/10.0, Coimbatore, India

AWARDS

5th Place, HackCovid 2021 Hackathon – built the website for a frontline-worker mental-health platform (team of 4).